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1. Introduction

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THE GENERALIZED ASSIGNMENT PROBLEM: VALID INEQUALITIES AND FACETS

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Three classes of valid inequalities based upon multiple knapsack constraints are derived for the generalized assignment problem. General properties of the facet defining inequalities are discussed and, for a special case, the convex hull is completely characterized. In addition, we prove that a basic fractional solution to the linear programming relaxation can be eliminated by a facet defining inequality associated with an individual knapsack constraint.

Key words: Generalized assignment problem, knapsack problem, special ordered sets, integer polytope, facets.

1. Introduction

The generalized assignment problem (GAP) is that of finding an optimal assignment of agents to tasks. Unlike the classical assignment problem which provides a one to one pairing of agents and tasks, in the generalized assignment problem an agent may be assigned to multiple tasks; ensuring, however, that each task is performed exactly once. The operations research and management science literature abounds with applications of GAP such as machine scheduling, lump sum capital rationing, and computer networking [1]. In addition, Ross and Soland [27, 28] have shown that a large variety of facility location problems can be formulated as generalized assignment problems.

It is well known that the assignment problem can be solved in polynomial time and that the optimal solution to the linear programming relaxation will always yield zero-one solutions (see, for example, [18]). GAP, on the other hand, is an NP-complete problem [10]. This problem has been studied extensively from an algorithmic point of view (see, for example, [1, 8, 21, 27, 28, 29]). In general, the computational requirements of current solution methods, which ignore the structure of the polytope associated with the constraint set, tend to increase very quickly with only a modest increase in the size of the problem.

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It has been demonstrated that polyhedral based algorithms are particularly geared toward the solution of large-scale problems which are typically encountered in practice. We note, for example, recent large-scale computational studies by [25] for the optimal solution of the traveling salesman problem; [14, 15] for the optimal triangularity of input-output matrices; and [6, 17] for the solution of large unstructured zero–one problems. Our ultimate goal is to derive a polyhedral based algorithm for solving large-scale generalized assignment problems. Since the facial structure of the polytope associated with GAP has not been previously studied, our present research efforts are directed toward this particular aspect of the polyhedral approach. In this paper we present various classes of valid inequalities and study properties of facet defining inequalities for the polytope associated with GAP. In [13] we derive a class of facet defining inequalities based upon $(1, k)$-configurations. Additional classes of facet defining inequalities are derived in [12] and in the first author's dissertation [11] where many of the results presented herein appear. The results we derive apply to closely related problems such as the one dimensional bin packing problem [9, 10], and particular scheduling problems on identical processors [19]. However, we do not consider these problems specifically in this paper.

In Section 2 the formulation of GAP is presented; we derive an equivalent formulation, denoted as P_{MN}, which is the specific problem we study. Definitions and notation are presented in Section 3. In Section 4 three classes of valid inequalities are derived. General properties of facets are derived in Section 5. We discuss a special case in which the convex hull of 0–1 solutions is completely characterized. In addition, we show that the facets of the polytope associated with certain subsets of the constraints are also facets of the polytope associated with P_{MN}. Section 6 is devoted to the linear programming relaxation of P_{MN}, denoted as LP_{MN}. Characteristics of basic fractional solutions to LP_{MN} are derived; and, subsequently, we prove that a basic fractional solution to LP_{MN} can be eliminated by a facet associated with an individual knapsack constraint.

2. Problem formulation

The generalized assignment problem (GAP) is defined as follows:

$$\text{minimize} \quad \sum_{i=1}^{m} \sum_{j=1}^{n} c'_{ij} x_{ij}$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} x_{ij} \leq b_i, \quad i \in M, \tag{1}$$

$$\sum_{i=1}^{m} x_{ij} = 1, \quad j \in N, \tag{2}$$

$$x_{ij} \in \{0, 1\}, \quad i \in M, j \in N, \tag{3}$$

where $M = \{1, \ldots, m\}$, $|M| \geq 2$, and $N = \{1, \ldots, n\}$.

We assume throughout th
(1) are, henceforth, referred
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We transform GAP into a
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variables can be dropped. Fi
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$$\text{maximize} \quad \sum_{i=1}^{m} \sum_{j=1}^{n} c$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} x_{i}$$

$$\sum_{i=1}^{m} x_{ij} \leq$$

$$x_{ij} \in \{0,$$

3. Definitions and notation

Denote as K_{MN} the convex
P_{MN}, that is

$$K_{MN} = \mathrm{conv}\{x \in \mathbb{R}^m$$

The zero vector and the n
solutions for P_{MN}; hence we

Proposition 3.1. *K_{MN} has di*

A *facet* is a maximal pro
The inequality $\pi x \leq \pi_0$ is a
(i) $x \in K_{MN}$ implies $\pi x \leq$
(ii) there exist exactly m
$\pi x^i = \pi_0$, $i = 1, \ldots, mn$.

We assume throughout the paper that GAP has a feasible solution. Constraints (1) are, henceforth, referred to as knapsack constraints. We assume that a_{ij}, $i \in M$, $j \in N$ are nonnegative rationals, and b_i, $i \in M$ are positive rationals, and we restrict our attention to problems such that $a_{ij} \leq b_i$ for, otherwise, the corresponding variable x_{ij} would necessarily equal zero. We make no assumption regarding the value of $\sum_{j \in N} a_{ij}$. If for some i, $\sum_{j \in N} a_{ij} \leq b_i$, then that knapsack constraint can be eliminated; however, the corresponding variables cannot be eliminated from the problem due to the presence of the special ordered set (SOS) constraints (2).

We transform GAP into an equivalent packing problem with all inequality constraints. This transformation, similar to that of [21], is done by introducing an artificial variable y_j for each equation (2) with a cost $\theta_j > \sum_{j=1}^{n} c_j^*$ where $c_j^* = \max c'_{ij}$, $i \in M$. This guarantees that these artificial variables will not appear in the optimal solution. By substituting $y_j = 1 - \sum_{i=1}^{m} x_{ij}$ in the objective function, these artificial variables can be dropped. Finally, letting $c_{ij} = \theta_j - c'_{ij}$, this transformation yields the following equivalent formulation, denoted as P_{MN}.

$$\text{maximize} \quad \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} - \sum_{j=1}^{n} \theta_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} x_{ij} \leq b_i, \quad i \in M, \tag{4}$$

$$\sum_{i=1}^{m} x_{ij} \leq 1, \quad j \in N, \tag{5}$$

$$x_{ij} \in \{0, 1\}, \quad i \in M, j \in N. \tag{6}$$

3. Definitions and notation

Denote as K_{MN} the convex hull of 0–1 vertices of the polytope associated with P_{MN}, that is

$$K_{MN} = \mathrm{conv}\{x \in \mathbb{R}^{mn} \mid x \text{ satisfies (4)–(6)}\}.$$

The zero vector and the mn unit vectors are affinely independent and valid 0–1 solutions for P_{MN}; hence we have the following:

Proposition 3.1. *K_{MN} has dimension mn.* □

A *facet* is a maximal proper face; or, equivalently, a face of dimension $mn - 1$. The inequality $\pi x \leq \pi_0$ is a facet defining inequality if

(i) $x \in K_{MN}$ implies $\pi x \leq \pi_0$,

(ii) there exist exactly mn affinely independent vertices x^i of K_{MN} satisfying $\pi x^i = \pi_0$, $i = 1, \ldots, mn$.

In general, affine independence does not imply linear independence; however, for facets such that $\pi_0 > 0$, it is well known that the two notions are equivalent. Since K_{MN} is a full dimensional polytope there exists a set of facets uniquely defined up to multiplication by a positive scalar. Henceforth, the facet defining inequalities themselves are referred to as the facets of K_{MN}.

The following definitions apply to generalized packing problems; however, given the special structure of our problem, we define these index configurations for an arbitrary knapsack constraint. For ease of notation the first subscript is dropped. Let

$$P_I = \operatorname{conv}\left\{ x \in \mathbb{R}^n \mid \sum_{j \in N} a_j \leq b,\ x_j \in \{0, 1\}, j \in N \right\}.$$

A subset $S \subseteq N$ is a *cover* if $\sum_{j \in S} a_j > b$. A subset $S \subseteq N$ with $|S| = s$ is an (s, k)-*cover* [23] if

(i) for all $K \subset S$, $|K| = k$, $\sum_{j \in K} a_j \leq b$;

(ii) for all $K_1 \subseteq S$, $|K_1| = k+1$, $\sum_{j \in K_1} a_j > b$.

If $k = s-1$, the (s, k)-cover is a *minimal cover*. The inequality $\sum_{j \in S} x_j \leq k$ obtained from an (s, k)-cover is a facet of the convex hull P_I^* where

$$P_I^* = P_I \cap \{x \in \mathbb{R}^n \mid x_j = 0, j \in N \backslash S\}.$$

A set $N' \cup \{z\}$ is a $(1, k)$-*configuration* [24] if $N' \subset N$, $|N'| = n'$ and $z \in N \backslash N'$ are such that

(i) $\sum_{j \in N'} a_j \leq b$,

(ii) $K \cup \{z\}$ is a minimal cover for every $K \subseteq N'$ with $|K| = k$, where k is any integer satisfying $2 \leq k \leq n'$.

Let

$$P_I^* = P_I \cap \{x \in \mathbb{R}^n \mid x_j = 0, j \in N \backslash (N' \cup \{z\})\}.$$

The facets of P_I^* are given by

$$(r-k+1)x_z + \sum_{j \in T(r)} x_j \leq r,$$

where $T(r) \subseteq N'$ is any subset of cardinality r, and r is any integer satisfying $k \leq r \leq n'$. Note that the $(1, k)$-configuration gives rise to $\sum_{r=k}^{n'} \binom{n'}{r}$ distinct facets of P_I^*. When $r = k$ we obtain minimal cover inequalities. Note that if $k = n'$ the $(1, k)$-configuration is a minimal cover.

Denote as P_{WH} arbitrary subproblems of P_{MN}, where $W \subseteq M$, $|W| = w$ and $H \subseteq N$, $|H| = h$. We assume, without loss of generality, that when dealing with subproblems of P_{MN} the knapsack constraints are numbered $i = 1, \ldots, w$. The problem P_{MN} is represented as a complete bipartite graph $G = (V, E)$ where $V = \{M, N\}$ is the node set and E is the edge set. Nodes $i \in M$ correspond to the m knapsack constraints and nodes $j \in N$ correspond to the n SOS constraints (5). The edge connecting a node $i \in M$ and a node $j \in N$ is denoted as (i, j). Similarly, P_{WH} can be represented as a complete bipartite graph.

We normally define facets
lifting procedure to obtain
problem. See, for example,
procedures; and [3, 26, 32, 3
Consequently, we introduce
convex hulls which are repr
$y \leq wh$ be the total number of
The convex hull in $\mathbb{R}^y$ is der
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4. Classes of valid inequaliti

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4.1. Cycle inequalities

Definition 4.1. Let $G' = (V$
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n_i; hence, $h = \sum_{i \in W} (n_i - 1)$.
$2p + 1 \geq 1$. For each node i
(i, f_i) and (i, g_i) correspond

Proposition 4.1. *Consider th*

(i) *$i \in Q$ implies N_i is a*

We normally define facets of lower dimensional polytopes and then employ a lifting procedure to obtain facets of the polytope associated with the original problem. See, for example, [22, 23, 30, 31] for discussions of sequential lifting procedures; and [3, 26, 32, 33] for discussions of simultaneous lifting procedures. Consequently, we introduce notation to describe subproblems and their associated convex hulls which are represented on possibly incomplete bipartite graphs. Let $y \leq wh$ be the total number of variables in the subproblem which we denote as P^y_{WH}. The convex hull in $\mathbb{R}^y$ is denoted as K^y_{WH}. This notation indicates that the graph is possibly incomplete but that at least one variable from each of the w knapsack constraints is present and at least one variable from each of the h SOS constraints is present. It is easy to verify that this polytope has dimension y.

4. Classes of valid inequalities

Valid inequalities for individual knapsack constraints are, obviously, valid for K_{MN}; and, in the next section we show that facets associated with the individual knapsack constraints are also facets of K_{MN}. We denote these inequalities as *individual* valid inequalities or *individual* facets. In this section, we restrict our attention to valid inequalities based upon more than one knapsack constraint; that is, *joint* valid inequalities.

Three classes of valid inequalities are derived. The first class is designated as cycle inequalities. The graph corresponding to the support (non-zero coefficients) of the inequalities can be represented on a connected bipartite graph containing exactly one even cycle. Generalized cover inequalities and $(1, k)$-configuration inequalities represent the second and third class of valid inequalities, respectively. The graph corresponding to the support of these latter two inequalities is connected and may or may not contain cycles. In this section we show that the inequalities are valid for a lower dimensional polytope K^y_{WH}, but, since all of the coefficients are nonnegative, the inequalities are obviously valid for K_{MN} as well.

4.1. Cycle inequalities

Definition 4.1. Let $G' = (V', E')$ be a subgraph of $G = (V, E)$ induced by $V' = \{W, H\}$, where $W \subseteq M$, $H \subseteq N$, and $E' \subset E$. Let $|W| = w$, $|H| = h$, $|E'| = y$. The subgraph G' has exactly one cycle and this cycle contains all nodes $i \in W$. Note that this implies that $h \geq w$ and $y = w + h$. Let $E'_c \subset E'$ be the set of edges on the cycle. Corresponding to each node $i \in W$ denote $N_i = \{j \in N \mid (i, j) \in E'\}$ where $|N_i| = n_i$; hence, $h = \sum_{i \in W} (n_i - 1)$. Suppose $W = P \cup Q$ where $P \cap Q = \emptyset$, $|Q| = q \geq 0$, $|P| = 2p + 1 \geq 1$. For each node $i \in W$ designate the two arcs on the cycle incident to i as (i, f_i) and (i, g_i) corresponding to x_{if_i} and x_{ig_i} where $a_{if_i} \leq a_{ig_i}$.

Proposition 4.1. *Consider the graph G' of Definition 4.1. If*

(i) *$i \in Q$ implies N_i is a cover for knapsack i,*

(ii) *$i \in P$ implies $N_i \backslash g_i$ is a cover for knapsack i;*

then

$$\sum_{(i,j) \in E'} x_{ij} \leq h - p - 1 \tag{7}$$

is a valid inequality for K^{ν}_{WH}.

Proof. Condition (i) implies that

$$\sum_{j \in N_i} x_{ij} \leq n_i - 1, \quad i \in Q. \tag{8}$$

Condition (ii) implies that

$$\sum_{j \in N_i \backslash g_i} x_{ij} \leq n_i - 2, \quad i \in P, \tag{9}$$

and, since $a_{if_i} \leq a_{ig_i}$, we have

$$\sum_{j \in N_i \backslash f_i} x_{ij} \leq n_i - 2, \quad i \in P. \tag{10}$$

In addition, due to the SOS constraints, we have the w cycle inequalities

$$x_{ij} + x_{kj} \leq 1, \quad j \in H, \ (i,j), (k,j) \in E'_c, \tag{11}$$

and the $\sum_{i \in Q} (n_i - 2)$ support bounds

$$x_{ij} \leq 1, \quad i \in Q, \ j \in N_i \backslash \{f_i, g_i\}. \tag{12}$$

Now, summing inequalities (8) to (12) we get

$$\begin{aligned}
&\sum_{i \in Q} \sum_{j \in N_i} x_{ij} + \sum_{i \in P} \left(\sum_{j \in N_i \backslash g_i} x_{ij} + \sum_{j \in N_i \backslash f_i} x_{ij} \right) \\
&\quad + \sum_{j \in H} \left(\sum_{i | (i,j) \in E'_c} x_{ij} \right) + \sum_{i \in Q} \sum_{j \in N_i \backslash \{f_i, g_i\}} x_{ij} \\
&= 2 \sum_{(i,j) \in E'} x_{ij} \\
&\leq \sum_{i \in Q} (n_i - 1) + 2 \sum_{i \in P} (n_i - 2) + w + \sum_{i \in Q} (n_i - 2) \\
&= 2 \sum_{i \in Q} (n_i - 1) - q + 2 \sum_{i \in P} (n_i - 1) - 2(2p + 1) + w \\
&= 2 \sum_{i \in W} (n_i - 1) - (2p + 1) \\
&= 2h - 2p - 1.
\end{aligned}$$

Thus,

$$\sum_{(i,j) \in E'} x_{ij} \leq h - p - \tfrac{1}{2},$$

and, due to 0–1 integrality, inequality (7) is valid for K^{ν}_{WH}. □

Example 4.1. Let

$(a_{11}, a_{12}) = (1, 1),$

$(a_{22}, a_{23}, a_{25}) = (1,$

$(a_{33}, a_{34}, a_{36}) = (1,$

$(a_{41}, a_{44}, a_{47}) = (1,$

$(a_{55}, a_{56}, a_{57}) = (2,$

$W = \{1, 2, 3, 4\},$

$H = \{1, 2, 3, 4, 5, 6,$

$N_1 = \{1, 2\}$ is a cov

$N_3 = \{3, 4, 6\}$ and

$N_4 = \{1, 4, 7\}$ and

Thus

$$\sum_{(i,j) \in E'} x_{ij} \leq 5,$$

where E' are the edges in (
(13) cuts off the following f
$x_{34} = x_{36} = \frac{1}{2}$; $x_{41} = x_{44} = x_{47} =$
see [16], that none of the fa
will cut off this fractional s

4.2. Generalized cover inequ

In this section we derive a
the support of these inequa

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Fig. 1. Graph G' for Example 4.

Example 4.1. Let

$$(a_{11}, a_{12}) = (1, 1), \qquad b_1 = 1,$$
$$(a_{22}, a_{23}, a_{25}) = (1, 1, 2), \qquad b_2 = 2,$$
$$(a_{33}, a_{34}, a_{36}) = (1, 1, 2), \qquad b_3 = 2,$$
$$(a_{41}, a_{44}, a_{47}) = (1, 1, 2), \qquad b_4 = 2,$$
$$(a_{55}, a_{56}, a_{57}) = (2, 1, 1), \qquad b_5 = 2,$$
$$W = \{1, 2, 3, 4\}, \qquad Q = \{1\}, \qquad P = \{2, 3, 4\}, \qquad p = 1,$$
$$H = \{1, 2, 3, 4, 5, 6, 7\}, \qquad h = 7,$$
$$N_1 = \{1, 2\} \text{ is a cover}, \qquad N_2 = \{2, 3, 5\} \text{ and } N_2 \backslash 3 \text{ is a cover},$$
$$N_3 = \{3, 4, 6\} \text{ and } N_3 \backslash 4 \text{ is a cover},$$
$$N_4 = \{1, 4, 7\} \text{ and } N_4 \backslash 4 \text{ is a cover}.$$

Thus

$$\sum_{(i,j) \in E'} x_{ij} \leq 5, \tag{13}$$

where E' are the edges in G' of Definition 4.1, is a valid inequality. Furthermore, (13) cuts off the following fractional solution: $x_{11} = x_{12} = \frac{1}{2}$; $x_{22} = x_{23} = x_{25} = \frac{1}{2}$; $x_{33} = x_{34} = x_{36} = \frac{1}{2}$; $x_{41} = x_{44} = x_{47} = \frac{1}{2}$; $x_{55} = x_{56} = x_{57} = \frac{1}{2}$; $x_{ij} = 0$, otherwise. It is easy to verify, see [16], that none of the facets associated with the individual knapsack constraints will cut off this fractional solution. The graph G' is shown in Figure 1.

4.2. Generalized cover inequalities

In this section we derive a class of inequalities based upon covers. The graph of the support of these inequalities is connected.

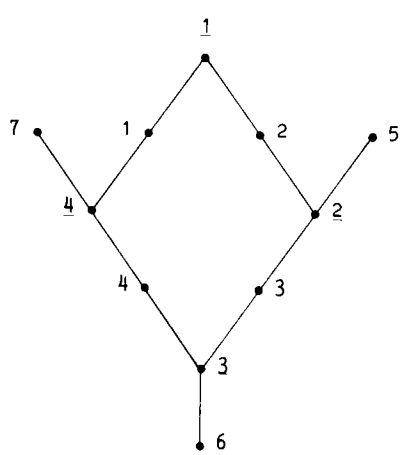

Fig. 1. Graph G' for Example 4.1. Nodes $i \in W = \{1, 2, 3, 4\}$ are underlined; nodes $j \in H = \{1, \ldots, 7\}$.

Proposition 4.2. *If*

(i) *for some* $p \in W$, $S \subset N$, $|S| = s$ *is such that for all* $K \subseteq S$ *with* $|K| = k+1$, $\sum_{j \in K} a_{pj} > b_p$,

(ii) *for each* $i \in W \backslash p$ *there exists a set* $N_i \subseteq N \backslash S$ *such that* $|N_i| = n_i$ *and a set* $S_i \neq \emptyset$ *with* $|S_i| = s_i$ *where*

$$S_i = \left\{ j \in S \,|\, \text{for all } R \subseteq \{j\} \cup N_i,\ |R| = r_i + 1,\ \sum_{j \in R} a_{ij} > b_i \right\};$$

then

$$\sum_{j \in S} x_{pj} + \sum_{i \in W \backslash p} \sum_{j \in S_i \cup N_i} x_{ij} \leq s + \sum_{i \in W \backslash p} r_i - \lceil (s-k)/s_q \rceil, \tag{14}$$

where $s_q = \max\{s_i \,|\, i \in W \backslash p\}$, *is a valid inequality for* K^v_{WH}, $H = S \cup \{\bigcup_{i \in W \backslash p} N_i\}$.

Proof. Condition (i) implies that

$$\sum_{j \in S} x_{pj} \leq k. \tag{15}$$

Condition (ii) implies that

$$x_{iu} + \sum_{j \in N_i} x_{ij} \leq r_i, \quad i \in W \backslash p,\ u \in S_i. \tag{16}$$

The SOS constraints imply that

$$\sum_{i \in W} x_{ij} \leq 1, \quad j \in S. \tag{17}$$

Multiplying (15) by $1/s_q$, (16) by $1/s_i$, and (17) by $(s_q - 1)/s_q$ and adding yields

$$\begin{aligned} &\frac{1}{s_q} \sum_{j \in S} x_{pj} + \sum_{i \in W \backslash p} \frac{1}{s_i} \left\{ \sum_{u \in S_i} \left(x_{iu} + \sum_{j \in N_i} x_{ij} \right) \right\} + \frac{s_q - 1}{s_q} \sum_{j \in S} \sum_{i \in W} x_{ij} \\ &\leq \frac{k}{s_q} + \sum_{i \in W \backslash p} r_i + \frac{(s_q - 1)s}{s_q} \\ &= \sum_{i \in W \backslash p} r_i + \frac{k + s_q s - s}{s_q} \\ &= \sum_{i \in W \backslash p} r_i + s - \frac{s-k}{s_q}. \end{aligned} \tag{18}$$

Note that the left hand side of (18) equals

$$\sum_{j \in S} x_{pj} + \sum_{i \in W \backslash p} \left\{ \frac{s_q + s_i(s_q - 1)}{s_i s_q} \sum_{j \in S_i} x_{ij} + \frac{s_q - 1}{s_q} \sum_{j \in S \backslash S_i} x_{ij} + \sum_{j \in N_i} x_{ij} \right\}.$$

Truncating the fractional coefficients, noting that

$$1 \leq (s_q + s_i(s_q - 1))/(s_i s_q) < 2,$$

it follows that

$$\sum_{j \in S} x_{pj} + \sum_{i \in W \backslash p} \sum_{j \in S_i \cup N}$$

Due to 0–1 integrality inequ

Example 4.2. Let

$$(a_{11}, \ldots, a_{15}) = (1,$$

$$(a_{21}, a_{22}, a_{26}, a_{27}) =$$

$$(a_{33}, a_{34}, a_{36}, a_{37}, a$$

$$(a_{45}, a_{47}, a_{48}) = (1,$$

$$W = \{1, 2, 3\}, \qquad p$$

$$N_2 = \{6, 7\}, \qquad S_2 =$$

$$N_3 = \{6, 7, 8\}, \qquad S$$

$$s_q = \max\{s_i \,|\, i \in W \backslash$$

Hence,

$$\sum_{j \in S} x_{1j} + \sum_{i=2}^{3} \sum_{j \in S_i \cup N_i}$$

is a valid inequality. Furthe
$x_{11} = x_{12} = x_{13} = x_{14} = x_{15} = \frac{2}{5}$;
$x_{45} = x_{47} = x_{48} = \frac{3}{5}$; $x_{ij} = 0$ oth
associated with the individ
solution.

4.3. (1, k)-Configuration ine

In this section we derive va
configuration in one of the
generalized cover inequalitie
an entire family of inequalit

In the following proposit
configuration for some arbit
sets $T(r) \subseteq N'$, $|T(r)| = r$,
$t = 1, \ldots, \binom{n'}{r}$. The index f_i
$\min\{a_{ij} \,|\, j \in T_t^r\}$. If the minin
one index the index is arbit

it follows that

$$\sum_{j\in S} x_{pj} + \sum_{i\in W\setminus p} \sum_{j\in S_i\cup N_i} x_{ij} \leq \sum_{i\in W\setminus p} r_i + s - \frac{s-k}{s_q}.$$

Due to 0–1 integrality inequality (14) is valid for $K^{\vee}_{WH}$. □

Example 4.2. Let

$$(a_{11},\ldots,a_{15}) = (1,1,1,1,1), \qquad b_1 = 2,$$

$$(a_{21}, a_{22}, a_{26}, a_{27}) = (1,1,2,2), \qquad b_2 = 2,$$

$$(a_{33}, a_{34}, a_{36}, a_{37}, a_{38}) = (1,1,2,2,2), \qquad b_3 = 4,$$

$$(a_{45}, a_{47}, a_{48}) = (1,1,1), \qquad b_4 = 2,$$

$$W = \{1,2,3\}, \qquad p = 1, \qquad S = \{1,2,3,4,5\}, \quad s = 5, \qquad k = 2,$$

$$N_2 = \{6,7\}, \qquad S_2 = \{1,2\}, \quad s_2 = 2, \quad r_2 = 1,$$

$$N_3 = \{6,7,8\}, \qquad S_3 = \{3,4\}, \quad s_3 = 2, \quad r_3 = 2,$$

$$s_q = \max\{s_i \mid i \in W\setminus p\} = 2, \qquad H = \{1,\ldots,8\}.$$

Hence,

$$\sum_{j\in S} x_{1j} + \sum_{i=2}^{3} \sum_{j\in S_i\cup N_i} x_{ij} \leq 5 + 3 - \lceil (5-2)/2 \rceil = 6 \tag{19}$$

is a valid inequality. Furthermore, (19) cuts off the following fractional solution: $x_{11} = x_{12} = x_{13} = x_{14} = x_{15} = \frac{2}{5}$; $x_{21} = x_{22} = \frac{3}{5}$, $x_{26} = \frac{2}{5}$; $x_{33} = x_{34} = x_{36} = \frac{3}{5}$, $x_{37} = x_{38} = \frac{2}{5}$; $x_{45} = x_{47} = x_{48} = \frac{3}{5}$; $x_{ij} = 0$ otherwise. It is easy to verify that none of the facets associated with the individual knapsack constraints will cut off this fractional solution.

4.3. (1, k)-Configuration inequalities

In this section we derive valid inequalities based upon the existence of a $(1,k)$-configuration in one of the knapsack constraints. Unlike the situation with the generalized cover inequalities where just one inequality is generated, with this class an entire family of inequalities is generated.

In the following proposition we assume that $N'\cup\{z\}$ with $|N'| = n'$ is a $(1,k)$-configuration for some arbitrary knapsack p. For specific r, $k \leq r \leq n'$ there exist $\binom{n'}{r}$ sets $T(r) \subseteq N'$, $|T(r)| = r$, which are arbitrarily numbered and denoted as T^r_t, $t = 1,\ldots,\binom{n'}{r}$. The index f_i is such that $a_{if_i(r,t)} = \min\{a_{iz}, a_{id_i(r,t)}\}$ where $a_{id_i(r,t)} = \min\{a_{ij} \mid j \in T^r_t\}$. If the minimum in the above definitions is attained for more than one index the index is arbitrarily selected.

Proposition 4.3. *Suppose*

(i) $N' \cup \{z\}$, *where* $N' \subset N$, $z \in N \backslash N'$ *is a* $(1, k)$*-configuration for knapsack* p,

(ii) *for some* r, $k \leq r \leq n'$, *and some* t, $1 \leq t \leq \binom{n'}{r}$, $i \in W \backslash p$ *implies there exists a set* $N_i \subseteq N \backslash (N' \cup \{z\})$ *such that for all* $R \subseteq \{f_i(r, t)\} \cup N_i$, $|R| = r_i + 1$, $\sum_{j \in R} a_{ij} > b_i$; *then*

$$(r-k+1)x_{pz} + \sum_{j \in T_t^r} x_{pj} + \sum_{i \in W \backslash p} \sum_{j \in N_i(z,r,t)} x_{ij} \leq r + \sum_{i \in W \backslash p} r_i \tag{20}$$

is a valid inequality for K^v_{WH} *where* $N_i(z, r, t) = N_i \cup \{z\} \cup \{d_i(r, t)\}$ *and* $H = \{z\} \cup N' \cup \{\bigcup_{i \in W \backslash p} N_i\}$.

Proof. We show that (20) is a valid inequality for an arbitrary r and t. Suppose that $\bar{x}$ is a feasible solution to P^v_{WH}. We will show that $\bar{x}$ satisfies (20). There are two cases to consider depending on the value of $\bar{x}_{pz}$.

Case 1. $\bar{x}_{pz} = 0$. Due to the SOS constraints

$$\sum_{i \in W} \sum_{j \in T_t^r} \bar{x}_{ij} \leq r.$$

Since $a_{iz} \geq a_{if_i(r,t)}$, condition (ii) implies that

$$\sum_{j \in N_i \cup \{z\}} \bar{x}_{ij} \leq r_i, \quad i \in W \backslash p,$$

which implies that $\bar{x}$ satisfies (20).

Case 2. $\bar{x}_{pz} = 1$. We show that the sum of the remaining variables can never exceed

$$r + \sum_{i \in W \backslash p} r_i - (r-k+1) = \sum_{i \in W \backslash p} r_i + k - 1.$$

Since $N' \cup \{z\}$ is a $(1, k)$-configuration,

$$\sum_{j \in T_t^r} \bar{x}_{pj} \leq k - 1.$$

Since $a_{id_i(r,t)} \geq a_{if_i(r,t)}$, condition (ii) implies that

$$\sum_{j \in N_i \cup \{d_i(r,t)\}} \bar{x}_{ij} \leq r_i, \quad i \in W \backslash p.$$

Due to the SOS constraints, $\bar{x}_{iz} = 0$, $i \in W \backslash p$; hence, $\bar{x}$ satisfies (20) in this case also. □

If condition (ii) holds for every r, $k \leq r \leq n'$ and every t, $1 \leq t \leq \binom{n'}{r}$, then (20) will yield $\sum_{r=k}^{n'} \binom{n'}{r}$ valid inequalities.

Example 4.3. Let

$(a_{11}, \ldots, a_{14}) = (3,$

$(a_{21}, \ldots, a_{25}) = (1,$

$(a_{31}, \ldots, a_{36}) = (1,$

$(a_{45}, a_{46}) = (1, 1),$

$p = 1, \quad z = 1,$

$W = \{1, 2, 3\}, \quad N$

Then, since condition (ii) h

$(r-1)x_{11} + \sum_{j \in T_t^r} x_{1j} +$

are valid inequalities for $r =$
$r = 3$ the one set T_t^r is $T_1^3 =$

$2x_{11} + \sum_{j \in T_t^r} x_{1j} + x_{21}$

Furthermore, (22) cuts off th
$x_{21} = x_{23} = x_{25} = \frac{1}{2}$; $x_{32} = x_{35} =$
of the facets associated wit
fractional solution. In additi
yields the valid inequality

$x_{11} + x_{13} + x_{14} + x_{21}$

which also cuts off this frac

5. Facets

In this section general prop
facets and then consider a
together with the trivial facet
we focus on the general pro
from combinations of knap
facets of K^v_{WH} and facets of
with individual knapsack co
tions are easy to prove.

Proposition 5.1. *All nonneg*
K_{MN}. □

Proposition 5.2. *All SOS co*

The inequalities in the ab
is shown by Hammer, John

Example 4.3. Let

$$(a_{11},\ldots,a_{14})=(3,1,1,1),\qquad b_1=4,$$
$$(a_{21},\ldots,a_{25})=(1,2,1,2,2),\qquad b_2=2,$$
$$(a_{31},\ldots,a_{36})=(1,1,1,1,1,1),\qquad b_3=1,$$
$$(a_{45},a_{46})=(1,1),\qquad b_4=1,$$
$$p=1,\qquad z=1,\qquad N'=\{2,3,4\},\quad n'=3,\qquad k=2,$$
$$W=\{1,2,3\},\qquad N_2=\{5\},\quad N_3=\{5,6\},\quad r_2=r_3=1,\qquad H=\{1,\ldots,6\}.$$

Then, since condition (ii) holds for each possible value of r and t,

$$(r-1)x_{11}+\sum_{j\in T_t^r} x_{1j}+\sum_{i\in W\setminus p}\ \sum_{j\in N_i(1,r,t)} x_{ij}\leq r+2 \tag{21}$$

are valid inequalities for $r=2,3$. There are at least $\binom{3}{2}+\binom{3}{3}=4$ inequalities. When $r=3$ the one set T_t^r is $T_1^3=\{2,3,4\}$ and (21) is

$$2x_{11}+\sum_{j\in T_t^r} x_{1j}+x_{21}+x_{23}+x_{25}+x_{31}+x_{32}+x_{35}+x_{36}\leq 5. \tag{22}$$

Furthermore, (22) cuts off the following fractional solution: $x_{11}=x_{12}=x_{13}=\frac{1}{2}$, $x_{14}=1$; $x_{21}=x_{23}=x_{25}=\frac{1}{2}$; $x_{32}=x_{35}=\frac{1}{2}$; $x_{46}=1$; $x_{ij}=0$ otherwise. It is easy to verify that none of the facets associated with the individual knapsack constraints will cut off this fractional solution. In addition, now letting $W=\{1,2\}$, when $r=2$, the set $T_t^2=\{3,4\}$ yields the valid inequality

$$x_{11}+x_{13}+x_{14}+x_{21}+x_{23}+x_{25}\leq 3$$

which also cuts off this fractional solution.

5. Facets

In this section general properties of facets are derived. We first define the trivial facets and then consider a special case in which the *individual* nontrivial facets together with the trivial facets completely characterize the convex hull. Subsequently, we focus on the general properties of nontrivial *joint* facets; that is, facets derived from combinations of knapsack constraints. In addition, the relationship between facets of K_{WH}^{v} and facets of K_{MN} is studied and we show that the facets associated with individual knapsack constraints are facets of K_{MN}. The following two propositions are easy to prove.

Proposition 5.1. *All nonnegativity constraints* $x_{ij}\geq 0$, $i\in M$, $j\in N$, *are facets of* K_{MN}. □

Proposition 5.2. *All SOS constraints* (5) *are facets of* K_{MN}. □

The inequalities in the above two propositions are defined to be trivial facets. It is shown by Hammer, Johnson and Peled [16] that, since the constraint matrix is

nonnegative and all constraints are less than or equal to inequalities, all *nontrivial* facets are of the form $\sum_{i\in M}\sum_{j\in N}\pi_{ij}x_{ij}\leq\pi_0$, denoted as $\pi x\leq\pi_0$, with $\pi_{ij}\geq 0$ and $\pi_0>0$. The following lemma is employed in some of the proofs in this section. A related result appears in [2] for canonical facets of the knapsack polytope.

Lemma 5.1. *Assume that $\pi x\leq\pi_0$ is a facet of K_{MN}. If the facet is nontrivial, then for any $k\in N$ there exists a 0–1 solution x^* such that $\pi x^*=\pi_0$ and $\sum_{i\in M}x^*_{ik}=0$.*

Proof. If all solutions on the facet are such that $\sum_{i\in M}x_{ik}=1$, the facet must be a trivial constraint which is a contradiction. □

5.1. A special case

Proposition 5.3 in this section provides sufficient conditions for a variable to have a zero coefficient in every nontrivial facet of K_{MN}. These conditions typically arise when the knapsack coefficients consist of a set of large elements together with a set of relatively small elements. The following lemma is used in the proof of Proposition 5.3 and Theorem 6.1.

Lemma 5.2. *Let $T=\{t_1,\ldots,t_k\}\subset N$, $|T|\geq 1$ and $a_{t_i}\geq a_{t_{i+1}}$, $i=1,\ldots,k-1$. Assume $R\subset N$, $|R|\geq 1$, $R\cap T=\emptyset$. If*

(i) *$R\cup T$ is a cover,*

but

(ii) *$R\cup T\backslash t_k$ is not a cover;*

then there exists a minimal cover $S\cup T$, $S\subseteq R$, $|S|\geq 0$.

Proof. Let $R=\{j_1,\ldots,j_r\}$ where $a_{j_i}\geq a_{j_{i+1}}$, $i=1,\ldots,r-1$. Define $R_1=\{j\in R\,|\,a_j\geq a_{t_k}\}$, $|R_1|=r_1\geq 0$. Suppose $R_1\cup T$ is a cover. By condition (ii), $R_1\cup T\backslash t_k$ is not a cover; therefore, $R_1\cup T$ is a minimal cover and $S=R_1$. Now suppose that $R_1\cup T$ is *not* a cover. By condition (i) it follows that $R_1\neq R$. Define $R_2=\{j_{r_1+1},\ldots,j_{r_1+r_2}\}$, where $r_2\geq 0$, such that

$$\sum_{j\in R_1\cup T\cup R_2}a_j\leq b$$

and

$$\sum_{j\in R_1\cup T\cup R_2}a_j+a_{j_{r_1+r_2+1}}>b.$$

Note that this is always possible since $|R|\geq r_1+r_2+1$ due to condition (i). It follows that $S\cup T$ is a minimal cover where $S=R_1\cup R_2\cup\{j_{r_1+r_2+1}\}$. □

Proposition 5.3. *For a knapsack $p\in M$, suppose N is partitioned into two mutually disjoint sets N_{p1} and N_{p2}, one of them possibly empty, such that*

$$N_{p1}=\{j\in N\,|\,\textit{there exists a minimal cover for knapsack } p \textit{ containing } j\},$$

$$N_{p2}=N\backslash N_{p1}.$$

Then, in every nontrivial facet, $\sum_{i\in M}\sum_{j\in N}\pi_{ij}x_{ij}\leq\pi_0$, $\pi_{pj}=0$, $p\in M$, $j\in N_{p2}$.

Proof. Suppose that there ex

$$Q_{p2}=\{j\in N_{p2}\,|\,\pi_{pj}>$$

Since the facet is nontrivial, exists a 0–1 solution x^* on t

$$U^*_p=\{j\in N_{p1}\,|\,x^*_{pj}=$$

Clearly, U^*_p is not a cover otherwise, there exists a vali $x^{**}_{pk}=1$, satisfying $\pi x^{**}>\pi$ that a subset of U^*_p together by the definition of N_{p2}. H $|Q_{p2}|=0$. □

Remark 5.1. If for any knap $\pi_{pj}=0$, $j\in N$, in every nont

Proposition 5.4. *For $i=1,\ldots$ 5.3. Suppose N_{i1} is an $(n_{i1},$*

$$\sum_{j\in N_{i1}}x_{ij}\leq k_i,\quad i\in I$$

together with the trivial face

Proof. From Proposition 5 $j\in N_{i2}$. The following show by the single inequality (2

$$K_{i1}=\text{conv}\left\{x\in\mathbb{R}\right.$$

and

$$K_{i2}=\text{conv}\left\{x\in\mathbb{R}'\right.$$

All vertices of K_{i2} are inte hence, $K_{i2}\subseteq K_{i1}$. Assume Then at least k_i+1 variabl an (n_{i1},k_i)-cover. Hence, be replaced by inequality (tion polytope which is tot

Proposition 5.4 implies individual facets are the convex hull. Since our ain is the facial structure of N_{i1} is *not* an (n_{i1},k_i)-cov

Proof. Suppose that there exists a nontrivial facet such that $|Q_{p2}|>0$, where

$$Q_{p2}=\{j\in N_{p2}\,|\,\pi_{pj}>0\}.$$

Since the facet is nontrivial, it follows from Lemma 5.1 that for any $k\in Q_{p2}$ there exists a 0–1 solution x^* on the facet such that $\sum_{i\in M} x^*_{ik}=0$. Let

$$U^*_p=\{j\in N_{p1}\,|\,x^*_{pj}=1\}.$$

Clearly, U^*_p is not a cover for knapsack p. However, $U^*_p\cup\{k\}$ must be a cover; otherwise, there exists a valid 0–1 solution x^{**}, which is identical to x^* except that $x^{**}_{pk}=1$, satisfying $\pi x^{**}>\pi_0$. It follows from Lemma 5.2, with $R=U^*_p$ and $T=\{k\}$, that a subset of U^*_p together with $\{k\}$ will form a minimal cover. But this is impossible by the definition of N_{p2}. Hence, it must be the case that for any nontrivial facet $|Q_{p2}|=0$. □

Remark 5.1. If for any knapsack $p\in M$, $\sum_{j\in N} a_{pj}\leq b_p$, it follows that $N_{p1}=\emptyset$; hence, $\pi_{pj}=0$, $j\in N$, in every nontrivial facet of K_{MN}.

Proposition 5.4. *For $i=1,\ldots,m$ let N_{i1}, with $|N_{i1}|=n_{i1}$, be as defined in Proposition 5.3. Suppose N_{i1} is an (n_{i1},k_i)-cover. Then*

$$\sum_{j\in N_{i1}} x_{ij}\leq k_i,\quad i\in M, \tag{23}$$

together with the trivial facets define K_{MN}.

Proof. From Proposition 5.3 it follows that in every nontrivial facet $\pi_{ij}=0$, $i\in M$, $j\in N_{i2}$. The following shows that for any i the knapsack constraint can be replaced by the single inequality (23) for i. Define

$$K_{i1}=\operatorname{conv}\left\{x\in\mathbb{R}^n\,|\,\sum_{j\in N} a_{ij}x_{ij}\leq b_i,\ x_{ij}\in\{0,1\},j\in N\right\}$$

and

$$K_{i2}=\operatorname{conv}\left\{x\in\mathbb{R}^n\,|\,\sum_{j\in N_{i1}} x_{ij}\leq k_i,\ 0\leq x_{ij}\leq 1,j\in N\right\}.$$

All vertices of K_{i2} are integer and are feasible for K_{i1} since N_{i1} is an (n_{i1},k_i)-cover; hence, $K_{i2}\subseteq K_{i1}$. Assume that there exists a 0–1 solution $\bar{x}\in K_{i1}$ such that $\bar{x}\notin K_{i2}$. Then at least k_i+1 variables $\bar{x}_{ij}=1$, $j\in N_{i1}$. But this contradicts the fact that N_{i1} is an (n_{i1},k_i)-cover. Hence, $K_{i1}=K_{i2}$. Consequently, each knapsack constraint may be replaced by inequality (23) and the resultant constraint set is that of the transportation polytope which is totally unimodular. □

Proposition 5.4 implies that if N_{i1} is a (n_{i1},k_i)-cover for each $i\in M$, then the individual facets are the only nontrivial facets which are required for defining the convex hull. Since our aim is to study *joint* facets of K_{MN}, the focus of our research is the facial structure of K_{MN} when at least one knapsack constraint is such that N_{i1} is *not* an (n_{i1},k_i)-cover.

5.2. Properties of nontrivial joint facets

In this section we study general properties of nontrivial joint facets of K^{v}_{WH}. It is easy to verify that Lemma 5.1, Proposition 5.3, and Proposition 5.4 are also true for facets of K^{v}_{WH}. Throughout this section we consider the nontrivial joint facet

$$\sum_{i\in W}\sum_{j\in N_i}\pi_{ij}x_{ij}\leq\pi_0, \tag{24}$$

where

$$N_i=\{j\in N\mid \pi_{ij}>0\}$$

and $|N_i|=n_i$. As a consequence of Proposition 5.4 we have the following:

Proposition 5.5. *If* (24) *is a nontrivial joint facet of* K^{v}_{WH}, *then* N_i *is not an* (n_i, k_i)-*cover for some* i. □

Proposition 5.6. *Assume that* (24) *is a nontrivial facet of* K^{v}_{WH}. *Then*

$$\sum_{j\in N_i}a_{ij}>b_i$$

for all i.

Proof. Assume that $\sum_{j\in N_p}a_{pj}\leq b_p$ for some knapsack constraint p. Since the inequality is valid, $\pi x=\pi_0$ implies that $\sum_{i\in W}x_{ij}=1$ for each $j\in N_p$; consequently, by Lemma 5.1, (24) does not define a nontrivial facet. □

Remark 5.2. Note that the above proposition implies that $|N_i|\geq 2$ since $a_{ij}\leq b_i$.

Proposition 5.7. *In any nontrivial joint facet of* K^{v}_{WH}, $\pi_{ij}<\pi_0$, $i\in W$, $j\in N$.

Proof. Assume that $\pi_{pk}=\pi_0$ for some $p\in W$. Since we are considering a nontrivial joint facet there exists a $q\in W\backslash p$. Furthermore, since $|N_q|\geq 2$, there exists a $u\in N$, $u\neq k$, such that $\pi_{qu}>0$. Hence the 0-1 solution x^* defined by $x^*_{pk}=x^*_{qu}=1$, $x^*_{ij}=0$ otherwise, belongs to K^{v}_{WH}. But then $\pi x^*>\pi_0$, which is a contradiction. □

A consequence of Proposition 5.7 is that there are no unit vectors on a nontrivial joint facet. Hence we have the following:

Proposition 5.8. *The inequality* (24) *with* $\pi_{ij}\in\{0,1\}$, $\pi_0=1$ *is a facet of* K^{v}_{WH} *if and only if it is either*

(i) *an SOS constraint, or*

(ii) *an individual facet for some knapsack constraint.* □

The following theorem gives a necessary condition that must be satisfied by sets N_i, $i\in W$. For each $p\in M$ define

$$N^{(p)}=\bigcup_{i\in W\backslash p}N_i.$$

Theorem 5.1. *If the inequal*
$p\in W$,

$$|N_p\cap N^{(p)}|\geq 2.$$

Proof. For an arbitrary $p\in$
then there exists an index k

$$N_p\cap N^{(p)}=\{k\}.$$

Since (24) is a nontrivial f
that $\pi\bar{x}=\pi_0$ and $\bar{x}_{ik}=0$ for

$$\sum_{i\in W\backslash p}\sum_{j\in N_i}\pi_{ij}\bar{x}_{ij}=$$

and

$$\sum_{j\in N_p}\pi_{pj}\bar{x}_{pj}=\pi_{20}.$$

Necessarily, $\pi_{10}+\pi_{20}=\pi_0$.

Consider another 0-1 so
$\sum_{i\in W\backslash p}\sum_{j\in N_i}\pi_{ij}x^*_{ij}>\pi_{10}$ we
$x'_{ij}=x^*_{ij}$, $i\in W\backslash p$, $j\in N_i$; x'_{p}
since from (25) it follows t
$\sum_{i\in W}x_{ik}\leq 1$ which is clearl

On the other hand, if $\sum_i$
can construct a solution x'
$x'_{pj}=x^*_{pj}$, $j\in N_p$; $x'_{ij}=0$, oth
that $\sum_{i\in W\backslash p}\sum_{j\in N_i}\pi_{ij}x_{ij}=\pi_1$
of all 0-1 solutions on the
$|N_p\cap N^{(p)}|=0$ the same arg

As a consequence of The

Corollary 5.1. *In the bipartit*
joint facet there exists at lea

The following proposition
ing exactly one cycle.

Proposition 5.9. *If the graph*
contains exactly one cycle, th

Proof. Assume that all node
nodes $i\in W$ are on the cycle
that N_i is a (2, 1)-cover for
that (24) cannot be a nontri

Theorem 5.1. *If the inequality* (24) *is a nontrivial joint facet of* K^{y}_{WH}, *then for each* $p \in W$,

$$|N_p \cap N^{(p)}| \geq 2.$$

Proof. For an arbitrary $p \in W$ assume that $|N_p \cap N^{(p)}| < 2$. Suppose $|N_p \cap N^{(p)}| = 1$, then there exists an index k such that

$$N_p \cap N^{(p)} = \{k\}. \tag{25}$$

Since (24) is a nontrivial facet, there exists, by Lemma 5.1, a 0-1 solution $\bar{x}$ such that $\pi\bar{x} = \pi_0$ and $\bar{x}_{ik} = 0$ for all $i \in W$. Let

$$\sum_{i \in W \setminus p} \sum_{j \in N_i} \pi_{ij}\bar{x}_{ij} = \pi_{10}$$

and

$$\sum_{j \in N_p} \pi_{pj}\bar{x}_{pj} = \pi_{20}.$$

Necessarily, $\pi_{10} + \pi_{20} = \pi_0$.

Consider another 0-1 solution x^*, different from $\bar{x}$, which is on the facet. If $\sum_{i \in W \setminus p} \sum_{j \in N_i} \pi_{ij}x^*_{ij} > \pi_{10}$ we can construct a solution x' such that $\pi x' > \pi_0$ as follows: $x'_{ij} = x^*_{ij}$, $i \in W \setminus p$, $j \in N_i$; $x'_{pj} = \bar{x}_{pj}$, $j \in N_p$; $x'_{ij} = 0$, otherwise. The solution x' is valid since from (25) it follows that the only SOS constraint that must be considered is $\sum_{i \in W} x_{ik} \leq 1$ which is clearly satisfied.

On the other hand, if $\sum_{i \in W \setminus p} \sum_{j \in N_i} \pi_{ij}x^*_{ij} < \pi_{10}$, then $\sum_{j \in N_p} \pi_{pj}x^*_{pj} > \pi_{20}$ and we can construct a solution x' such that $\pi x' > \pi_0$ as follows: $x'_{ij} = \bar{x}_{ij}$, $i \in W \setminus p$, $j \in N_i$; $x'_{pj} = x^*_{pj}$, $j \in N_p$; $x'_{ij} = 0$, otherwise. Thus, all 0-1 solutions on the facet must be such that $\sum_{i \in W \setminus p} \sum_{j \in N_i} \pi_{ij}x_{ij} = \pi_{10}$ and $\sum_{j \in N_p} \pi_{pj}x_{pj} = \pi_{20}$. Hence, the rank of the matrix of all 0-1 solutions on the facet is strictly less than y and (24) is not a facet. If $|N_p \cap N^{(p)}| = 0$ the same argument holds. □

As a consequence of Theorem 5.1 we have the following corollary.

Corollary 5.1. *In the bipartite graph of the support (non-zero coefficients) of a nontrivial joint facet there exists at least one cycle.* □

The following proposition provides a necessary condition for joint facets containing exactly one cycle.

Proposition 5.9. *If the graph of the support of the nontrivial joint facet* (24) *of* K^{y}_{WH} *contains exactly one cycle, then there exists a node* $j \in N$ *which is not on the cycle.*

Proof. Assume that all nodes $j \in N$ are on the cycle. Suppose that, in addition, all nodes $i \in W$ are on the cycle. Proposition 5.6 and the assumption that $a_{ij} \leq b_i$ imply that N_i is a (2, 1)-cover for each $i \in W$. However, from Proposition 5.5 it follows that (24) cannot be a nontrivial joint facet. Hence, it must be the case that there

exists a node $p \in W$ which is not on the cycle. But, since all $j \in N$ are on the cycle and there is only one cycle, N_p is not a cover; hence, by Proposition 5.6, (24) cannot be a nontrivial facet. Since each of these possibilities produces a contradiction, it must be the case that there exists a node $j \in N$ which is not on the cycle. □

Remark 5.3. Note that for nontrivial facets containing multiple cycles, it is possible that all nodes $j \in N$ are on cycles.

5.3. *Facets of subproblems of* P_{MN}

The following proposition provides sufficient conditions for facets associated with subproblems to be facets of K_{MN}.

Proposition 5.10. *Given a nontrivial facet of* K^{y}_{WH} *with* $|W| \geq 1$, *if* $y = wn$, *the inequality is a facet of* K_{MN}.

Proof. Since the facet is nontrivial, for any $k \in N$ there exists at least one solution on the facet such that $\sum_{i \in W} x_{ik} = 0$. Therefore, when lifting x_{ik}, $i \notin W$ the lifting procedure will always yield a coefficient of zero. □

An important implication of this is the following corollary.

Corollary 5.2. *Nontrivial facets associated with individual knapsack constraints are facets of* K_{MN}. □

Note that Proposition 5.10 applies to facets of K_{WN} and not to facets of K_{WH}, $H \subset N$. In fact, Proposition 5.10 is, in general, not true if K_{WN} is replaced by K_{WH}. We contrast this to the plant location problem where facets associated with the polytope for a specific number of plants and a specific number of destinations remain facets if either or both the number of plants or destinations is increased. See [4, 5] for more details.

6. Linear programming relaxation

6.1. *Characteristics of basic fractional solutions to the linear programming relaxation of* P_{MN}

The linear programming relaxation of P_{MN}, denoted as LP_{MN}, is derived from P_{MN} by replacing (6) with

$$0 \leq x_{ij}, \quad i \in M, j \in N. \tag{26}$$

We assume that the θ_j values are such that (5) holds as strict equality in the optimal solution to LP_{MN}.

The relaxed problem, LP
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additional arc; thus, it is a c
[7, Chapter 21] or [20] for r

Consider the connected co
to LP_{MN} after removing
Necessarily, a forest of trees
all arcs in trees and 1-trees
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follows: We view a 1-tree as
considered as roots of the tr
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Proposition 6.1. *In any tre*
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Proof. In a tree, if a variabl
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Proposition 6.2. *Given a*
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than v. □

The relaxed problem, LP_{MN}, is a generalized network problem with inequality constraints, and it is well known that basic solutions for this class of problems can be represented as a forest consisting of trees and 1-trees. A 1-tree is a tree with one additional arc; thus, it is a connected component containing exactly one cycle, see [7, Chapter 21] or [20] for more details.

Consider the connected components of the graph obtained from a basic solution to LP_{MN} after removing arcs corresponding to degenerate basic variables. Necessarily, a forest of trees and 1-trees still remains. Henceforth, we assume that all arcs in trees and 1-trees correspond to positive variables only. It is well known that a tree will have at most one node whose slack variable is positive. If such a node, which is necessarily a node $i \in M$, exists, the tree is rooted at that particular node; otherwise, the tree is rooted at any node $i \in M$. An end node of a tree is a node which has only one edge incident to it. In a 1-tree there do not exist any nodes such that the slack variable is positive. An end node of a 1-tree can be defined as follows: We view a 1-tree as though it is a tree, such that all nodes on the cycle are considered as roots of the tree. An end node for this tree is a node which has only one edge incident to it. All such end nodes of this tree are end nodes of the 1-tree.

Proposition 6.1. *In any tree or 1-tree all variables x_{ij} are fractional except those corresponding to arcs incident to end nodes $j \in N$, which are equal to one.*

Proof. In a tree, if a variable associated with an arc which is not incident to an end node is equal to one, then , due to the SOS constraints, the tree would be disconnected which is impossible. Each end node corresponds to a node $j \in N$ since a node $i \in M$ at the end of a tree implies, by connectivity, that it is the end point of an arc associated with a fractional variable x_{ij}. This, in turn, implies that $a_{ij} > b_i$ which is a contradiction. Variables corresponding to arcs incident to end nodes are equal to one due to the SOS constraints. The same argument holds for 1-trees. □

Proposition 6.2. *Given a cycle containing $2v$ nodes, the sum of the variables corresponding to the arcs on the cycle is bounded by v, and is exactly equal to v if and only if the nodes $i \in M$ in the 1-tree are all on the cycle.*

Proof. There are v nodes $j \in N$ on the cycle; thus, due to the SOS constraints the maximum value for the sum of the variables corresponding to the arcs on the cycle is v. If the nodes $i \in M$ in the 1-tree are all on the cycle, then for any node $j \in N$, with adjacent nodes $f, g \in M$, $x_{fj} + x_{gj} = 1$; hence, the sum equals v. On the other hand, if there exists one node $p \in M$ which is not on the cycle, then p must necessarily be adjacent to some node $s \in N$ on the cycle. Suppose that nodes $f, g \in M$ are on the cycle and adjacent to node s. Then $x_{fs} + x_{gs} + x_{ps} \leq 1$. Since $x_{ps} > 0$, it follows that the sum of the variables corresponding to arcs on the cycle must be strictly less than v. □

Remark 6.1. Noting that the maximum number of positive components, including slack variables, in a basic feasible solution to LP_{MN} is $m+n$, it can be shown by a simple counting argument that the maximum number of fractional components is $\min(2m, 2n)$.

Remark 6.2. If $a_{ij} = a_j$, $i \in M$, the problem can be transformed into a transportation problem by setting $x_{ij} = t_{ij}/a_j$ for all $i \in M$, $j \in N$. If a_j and b_i are integer then in any basic solution t_{ij} will be integer.

6.2. Inequalities violated by basic fractional solutions to LP_{MN}

In this section we restrict our attention to basic fractional solutions such that at least one structural variable x_{ij} is fractional. We show that such a basic fractional solution to the relaxed problem LP_{MN} can be eliminated by an individual facet; more specifically, by a facet derived from a minimal cover inequality associated with a single knapsack. The proof relies on Lemma 5.2 which provides sufficient conditions for guaranteeing that a particular index set T is contained in a minimal cover.

Theorem 6.1. *Given a basic solution to* LP_{MN} *such that at least one structural variable* x_{ij} *is fractional, there exists a nontrivial individual facet of* K_{MN}, *derived from a minimal cover inequality, which is violated by this solution.*

Proof. Denote as $\bar{x}$ the vector of *nonzero* structural variables in an arbitrary basic fractional solution. We consider the following two mutually exclusive and exhaustive cases:

Case A. A tree containing a fractional arc or a 1-tree such that there exists a node $i \in M$ which is not on the cycle.

Case B. A 1-tree such that all nodes $i \in M$ are on the cycle.

Case A. For the connected components define:

$$W_c = \{i \in M \mid i \text{ is neither a root node nor on a cycle and } \textit{all} \text{ successor nodes are end nodes in some tree or 1-tree}\},$$

$$R_i = \{j \in N \mid j \text{ is an end node adjacent to node } i\}, \quad i \in W_c.$$

An example of these sets is exhibited in Figure 2.

Consider an arbitrary node $k \in W_c$ which is in a tree containing fractional arcs or a 1-tree. Assume that the predecessor node is $g \in N$. Note that $R_k \neq \emptyset$, $a_{kg}\bar{x}_{kg} + \sum_{j \in R_k} a_{kj}\bar{x}_{kj} = b_k$, and that $\bar{x}_{kj} = 1$, $j \in R_k$ and $0 < \bar{x}_{kg} < 1$. Thus, it follows that

$$\sum_{j \in R_k} a_{kj} < b_k \tag{27}$$

and

$$a_{kg} + \sum_{j \in R_k} a_{kj} > b_k. \tag{28}$$

Since $\{g\} \cup R_k$ is a cover,

$$x_{kg} + \sum_{j \in R_k} x_{kj} \leq r_k, \tag{29}$$

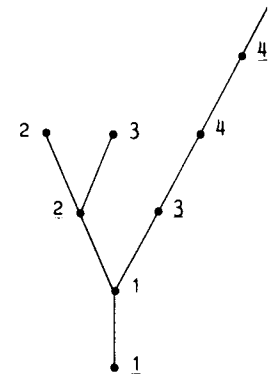

Fig. 2. Example demonstrating sets
nodes $j = 1, \ldots$

where $|R_k| = r_k$, is a valid in
it follows from Lemma 5.2,
$|S_k| = s_k$, such that $S_k \cup \{g\}$ i

$$x_{kg} + \sum_{j \in S_k} x_{kj} \leq s_k$$

is a violated valid inequality
$y = s_k + 1$.

Case B. For the connecte

$$W_c = \{i \in M \mid i \text{ is on}$$

$$R_i = \{j \in N \mid j \text{ is not}$$

An example of these sets is

This case is further broken
the nodes on the cycle adja

Fig. 3. Example demonstrating sets
nodes $j = 1, \ldots$

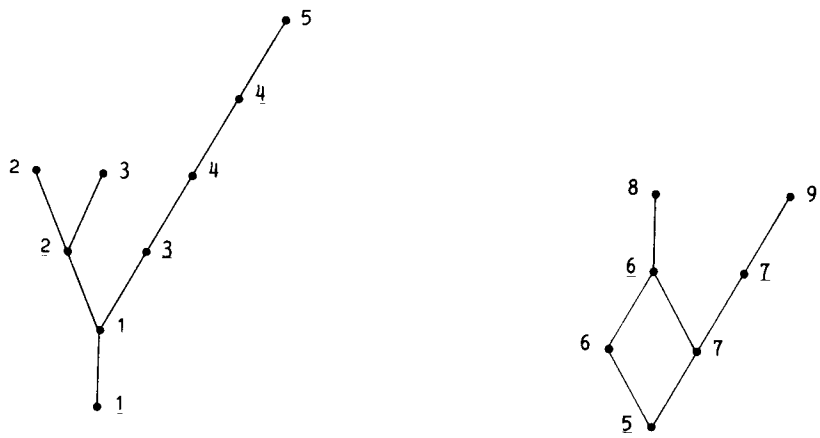

Fig. 2. Example demonstrating sets W_c and R_i of Theorem 6.1, Case A. Nodes $i = 1, \ldots, 7$ are underlined; nodes $j = 1, \ldots, 9$; $W_c = \{2, 4, 7\}$, $R_2 = \{2, 3\}$, $R_4 = \{5\}$, $R_7 = \{9\}$.

where $|R_k| = r_k$, is a valid inequality which is violated by $\bar{x}$. Due to (27) and (28), it follows from Lemma 5.2, with $T = \{g\}$, that there exists a set $S_k \subseteq R_k$, where $|S_k| = s_k$, such that $S_k \cup \{g\}$ is a minimal cover. Hence,

$$x_{kg} + \sum_{j \in S_k} x_{kj} \leq s_k \tag{30}$$

is a violated valid inequality and a facet of K^y_{WH}, with $W = \{k\}$, $H = \{g\} \cup S_k$, and $y = s_k + 1$.

Case B. For the connected components define:

$$W_c = \{i \in M \mid i \text{ is on a cycle}\},$$

$$R_i = \{j \in N \mid j \text{ is not on the cycle, but is adjacent to node } i\}, \quad i \in W_c.$$

An example of these sets is exhibited in Figure 3.

This case is further broken down into two subcases. Throughout, we assume that the nodes on the cycle adjacent to a node $i \in M$ are designated as $f_i, g_i \in N$ such

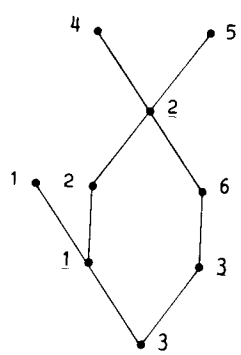

Fig. 3. Example demonstrating sets W_c and R_i of Theorem 6.1, Case B. Nodes $i = 1, \ldots, 3$ are underlined; nodes $j = 1, \ldots, 6$; $W_c = \{1, 2, 3\}$, $R_1 = \{1\}$, $R_2 = \{4, 5\}$, $R_3 = \emptyset$.

that $a_{if_i} \leq a_{ig_i}$. For ease of notation we omit the additional subscript since it should be clear that x_{if} refers to x_{if_i}, a_{if} refers to a_{if_i}.

Subcase (*i*). Suppose that there exists a node $k \in M$ on the cycle such that $\bar{x}_{kf} + \bar{x}_{kg} > 1$. If

$$a_{kf} + a_{kg} \leq b_k, \tag{31}$$

necessarily $R_k \neq \emptyset$ since $\bar{x}_{kf} < 1$ and $\bar{x}_{kg} < 1$. It follows that

$$a_{kf} + a_{kg} + \sum_{j \in R_k} a_{kj} > b_k, \tag{32}$$

since $a_{kf}\bar{x}_{kf} + a_{kg}\bar{x}_{kg} + \sum_{j \in R_k} a_{kj}\bar{x}_{kj} = b_k$. Note that since all nodes $i \in M$ are on the cycle, $\bar{x}_{kj} = 1$, $j \in R_k$; hence, R_k is not a cover. If $a_{kg} + \sum_{j \in R_k} a_{kj} > b_k$, from Lemma 5.2, with $T = \{g\}$ and $R = R_k$, there exists a set $S_k \subseteq R_k$ such that $S_k \cup \{g\}$ is a minimal cover. Hence, (30) is a violated valid inequality and a facet of K^{ν}_{WH}. On the other hand, if

$$a_{kg} + \sum_{j \in R_k} a_{kj} \leq b_k, \tag{33}$$

due to (32) and (33), we can apply Lemma 5.2 with $T = \{f, g\}$. Therefore, there exists a set $S_k \subseteq R_k$ such that $S_k \cup \{f, g\}$ is a minimal cover. Hence, since $\bar{x}_{kf} + \bar{x}_{kg} > 1$,

$$x_{kf} + x_{kg} + \sum_{j \in S_k} x_{kj} \leq s_k + 1 \tag{34}$$

is a violated valid inequality and a facet of K^{ν}_{WH}, with $W = \{k\}$, $H = \{f, g\} \cup S_k$, and $y = s_k + 2$. Note that $s_k \geq 1$ due to (31).

Suppose now that $a_{kf} + a_{kg} > b_k$. In this case the inequality (34) with $S_k = \emptyset$ is a violated inequality and a facet of K^{ν}_{WH}.

Subcase (*ii*). Suppose that no node $k \in M$ on the cycle satisfies $\bar{x}_{kf} + \bar{x}_{kg} > 1$. From Proposition 6.2 it follows that $\bar{x}_{if} + \bar{x}_{ig} = 1$ for all nodes $i \in M$ in the 1-tree. The following shows that since the linear programming solution is basic, there must exist some node $k \in M$ such that $R_k \neq \emptyset$ and $a_{kg} + \sum_{j \in R_k} a_{kj} > b_k$. Suppose that for each $i \in M$ in the 1-tree we have $a_{ig} + \sum_{j \in R_i} a_{ij} \leq b_i$, where R_i may possibly be empty. Since

$$b_i = a_{if}\bar{x}_{if} + a_{ig}\bar{x}_{ig} + \sum_{j \in R_i} a_{ij}\bar{x}_{ij} \leq a_{ig} + \sum_{j \in R_i} a_{ij} \leq b_i,$$

equality must hold throughout and $a_{if} = a_{ig}$. But, then each constraint corresponding to $i \in M$ in this 1-tree can be scaled such that $a_{if} = a_{ig} = 1$. Therefore, the columns corresponding to the variables in the 1-tree form a linearly dependent set due to the cycle and the solution cannot be basic. Hence, it must be the case that there exists at least one node $k \in M$ such that $a_{kg} + \sum_{j \in R_k} a_{kj} > b_k$; this in turn implies that $R_k \neq \emptyset$. Note that $\bar{x}_{kj} = 1$, $j \in R_k$; hence, R_k is not a cover. Now there exists a set $S_k \subseteq R_k$ such that (30), which is a facet of K^{ν}_{WH}, is violated by $\bar{x}$.

Therefore, for any basic fractional solution to LP_{MN} there exists a violated inequality of the form (30) or (34) associated with knapsack k. Each of these

inequalities is a minimal cov
facet associated with knapsac
the theorem follows. □

Acknowledgement

We would like to thank Ma
Several illuminating discussi
results.

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inequalities is a minimal cover inequality and can be lifted to obtain an individual facet associated with knapsack k, which, by Corollary 5.2, is a facet of K_{MN}. Hence, the theorem follows. □

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(1, k)-CONFIGURAT
ASSIGNMENT PRO

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A class of facet defining ineq
inequalities are based upon
configuration inequalities.

Key words: Generalized assig
polytope, facets, (1, k)-config

1. Introduction

In [3] we derive various class
problem (GAP). More restrict
these valid inequalities int
inequalities are particularly in
hull of 0–1 solutions.

In this paper we derive a
inequalities discussed in [3].
reader to Section 3 and Sectio
inequalities, which are specia
be found in [2] and [4].

2. (1, k)-Configuration facets

In this section we derive a cla
upon the joint valid inequali
$|N'| = n'$, is a (1, k)-configu
$k \leq r \leq n'$, there exist $\binom{n'}{r}$ sets

* Partial financial support unde
** Partial financial support unde